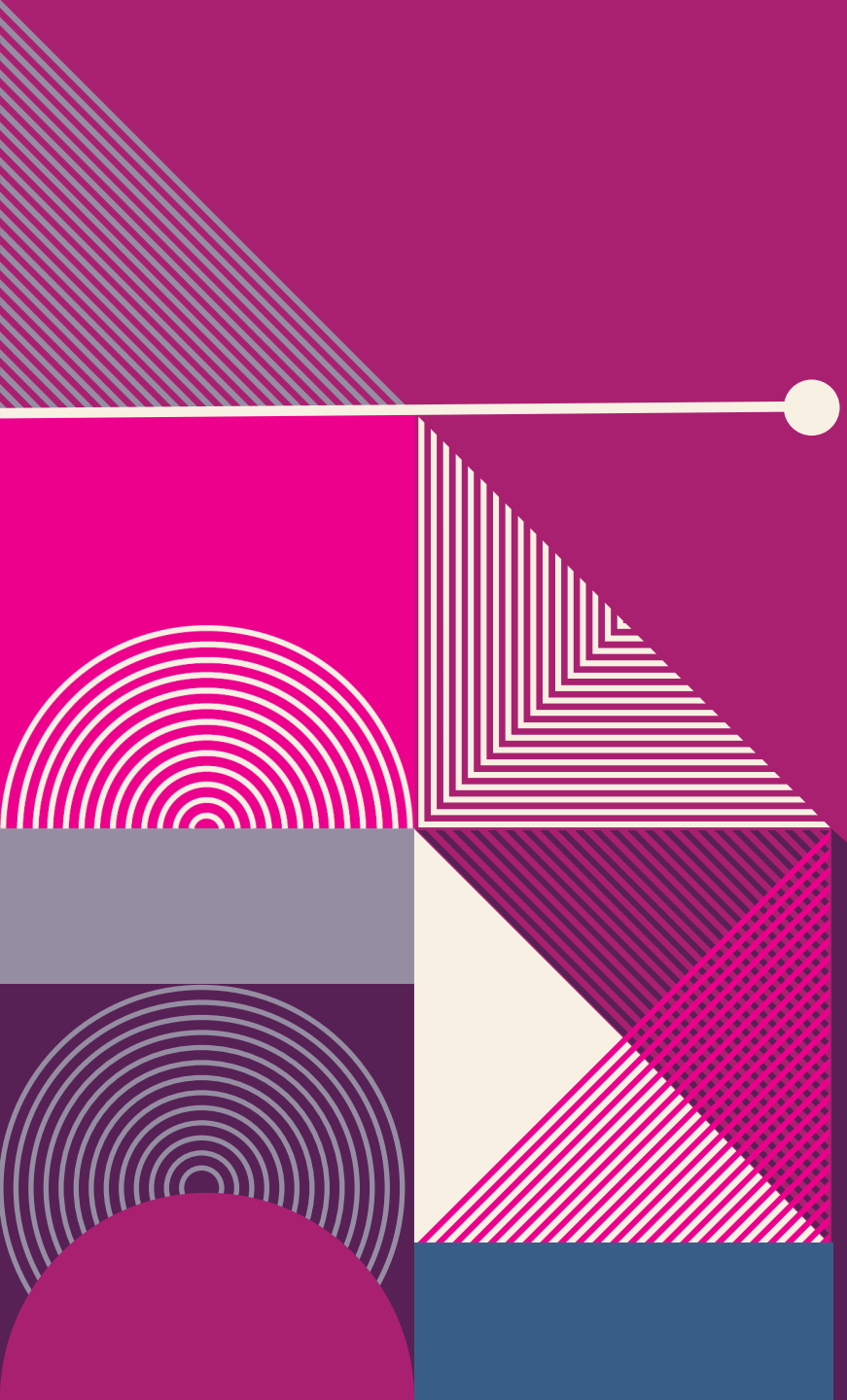
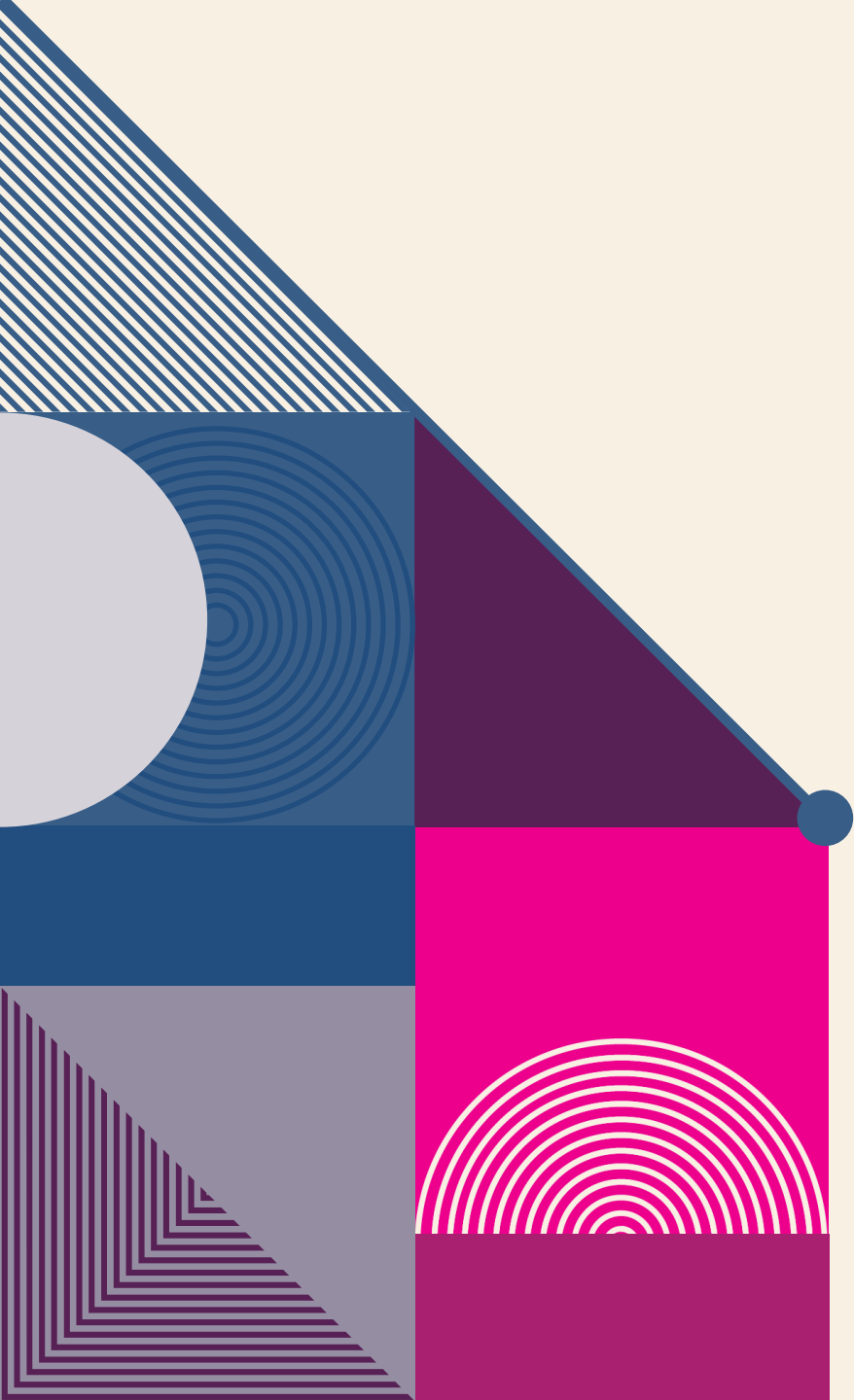




REASSESSING MODALITY IN FRENCH-BASED CREOLES: THE CASE OF DEONTIC AND EPISTEMIC MODALS.

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OVERVIEW

Haitian Creole and Martinican Creole

Rationale

Modality [MUST/CAN] and finiteness

Results and Discussion

MARTINICAN CREOLE AND HAITIAN CREOLE

- MC and HC are French-based creoles.
Lexically related, syntactically closely related.
- SVO languages
- Morphologically reduced languages : TMA
markers expressed as separate morphemes /
No morphological gender or number marking
on nouns and adjectives.

The mom liked her sons - La maman aimait ses fils

[MC] Manman-lan **té** enmen **sé** tchanmay-li **a**

[HC] Manman an **te** renmen pitit li **yo**



RATIONALE

- *"The process of pidginization involved loss of inflections in general and re-interpretation of some forms, including those of finite and nonfinite verbs. [...] So the verbs of Atlantic IEPC's [...] are neither finite nor nonfinite. [...] these languages have devised other strategies to convey what their lexifiers convey through the infinitive"* Mufwene & Dijkhoff 1989, p.325
- *"The finite/nonfinite distinction is disputable in Atlantic IEPC's [...] Either the facts that have been associated with putative distinction are not adequately accounted for or they can be described without unnecessarily complicating the grammar with it."* Mufwene & Dijkhoff 1989 p.325
- *"Il ne semble pas utile de distinguer dans les créoles français entre des formes verbales finies, donc tensées, et des formes non finies [...] Je conclurai qu'il n'existe ni infinitif ni participe dans les créoles français mais une grande diversité d'emplois transcatégoriels."* Véronique 2007

It does not seem useful to distinguish in French Creoles between finite verb forms, therefore tensed, and non-finite forms [...] I will conclude that there is neither infinitive nor participle in French Creoles but a great diversity of cross-categorical functions.

RATIONALE

1. Finiteness tied to inflectional morphology (finite vs. infinitive verb forms).
 - Infinitive verb form = nonfinite clauses || finite verb form = finite clauses
2. Creole languages have little to no inflectional morphology (Isolating languages like Mandarin Chinese)
 - Therefore, since no distinction finite/infinitive verb forms, this entails no distinction finite/nonfinite clauses
3. Do such languages make a finite/nonfinite distinction? How relevant is it? 2-fold approach:
 1. Demonstrate the viability of theories of finiteness for HC and MC
 2. Collect novel data to check for possible language change in progress, and analyze more recent versions of the language than those described 40+ years ago

THEORETICAL CONSIDERATIONS

- Bianchi (2003) : the syntax-semantics interface of finiteness
- Landau (2004) : setting subject licensing and tensedness apart
- Adger (2007) : Complement size matters
- Wurmbrand (2014, 2015) : Not all infinitives are equal
- Mittwoch (1990) : Bare infinitives

THEORETICAL FRAMEWORK

Finite/nonfinite distinction + fine-grained distinctions between complements

Indicative	Finite - full CP - pro/lexical DP - Independent tense <i>He says [that he will leave tomorrow / that he left yesterday]</i>
Subjunctive	Finite - full CP - pro/lexical DP - Dependent tense/future-oriented <i>It is necessary [that you be on time tomorrow / *that you were on time yesterday] Il est nécessaire [que tu sois à l'heure demain / *que tu aies été à l'heure hier]</i>
Partial control	Non-finite - full CP - PRO - Dependent tense/future-oriented <i>He prefers [to leave tomorrow / *to have left yesterday]</i>
ECM/raising	Non-finite - CP-truncated - Trace - Dependent tense/tensed <i>Anson believes Jill [*to call tomorrow / to have called yesterday]</i>
Exhaustive control	Non-finite - TP-truncated - Trace/PRO - Anaphoric tense/tenseless <i>*John begins [to leave tomorrow / to have left yesterday]</i>

Redefining tensedness and finiteness outside of morphological marking

- All finite clauses are tensed, but not all tensed clauses are finite
- A tensed clause is a clause with a non-anaphorical temporal specification
- A finite clause must be defined beyond its ability to license tense

GENERAL FINDINGS

Finite complement clauses					
Predicate types	Subject Type	Temporal Specification	Zero-marker Licensing	Negation Type	Complement Type
<i>Make</i>	Free	Independent	Yes	Sentential	Indicative
<i>Seem</i>	Free	Independent	Yes	Sentential	
<i>Fok</i>	Free	Future-oriented	No	Sentential	Subjunctive
<i>Want</i>	Free	Future-oriented	No	Sentential	

Nonfinite complement clauses					
<i>Want</i>	Non-coreferential	Future-oriented	No	vP	Partial control
<i>Seem</i>	Anaphorical	Dependent	No	vP	Raising
<i>Epistemic Modals</i>	Anaphorical	Dependent	No	vP	Raising
<i>Deontic Modals</i>	Anaphorical	Anaphorical	No	vP	Exhaustive control
<i>Make</i>	Non-coreferential	Anaphorical	No	vP	

Figure 1 – Description of the complementation systems of MC and HC

Note: The findings summarized in this figure suggest that the main properties that are absent in nonfinite complement clauses in HC and MC are 1) the free subject type, 2) the ability to license sentential negation, 3) the independent temporal specification on the complement clause and 4) its ability to license the zero-marker.

- Focus on MC pé/dwet and HC ka/dwe [MUST/CAN]
- MC and HC show evidence of a finite/nonfinite distinction
- Show evidence of complement size differences
- Allows us to better describe the behaviors observed in MC and HC
- Make cross-linguistic comparisons among Creole languages and among other isolating languages
- Depart from the Creole-lexifier systematic comparisons to explain Creole language change

EPISTEMIC MODALS AND ASPECT

Hu and Chen (2020) for Mandarin Chinese and Norwegian:

- Perfective aspect complements can only be interpreted as epistemic

When modals scope over an aspect marker, only an epistemic interpretation is available

1. « *The man must have disappeared* »
 - a. *It is likely that the man disappeared* (because the police cannot find him)
 - b. #*The man is/forced to have disappeared*
2. « *The man can have disappeared* »
 - a. *It is possible that the man disappeared* (because the police cannot find him)
 - b. #*The man is/was able to disappear*

This aligns with Cinque (1999) and his universal functional hierarchy and Magloire-Holly's (1982) claim in which epistemic modality scopes over T:

$$(10) \text{ Mod}_{\text{epistemic}} > \text{T(absolute)} > \text{Mod}_{\text{root}}^8$$

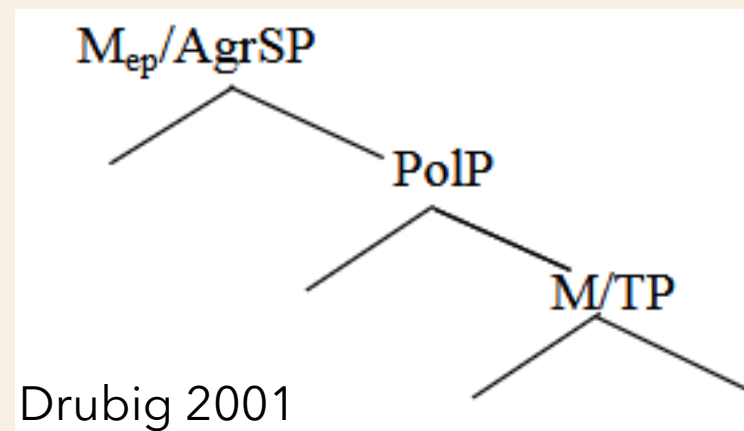
EPISTEMIC VS NONEPISTEMIC MODALITY

According to Enç (1996), nonepistemic modals are [+tense] and therefore license anaphorical predicates. They project at a lower position than epistemic modals.

McDowell (1987) identified syntactic behaviors separating M_{ep} (epistemic modals) and M modals (nonepistemic modals) also present in multiple modal constructions as found in AAE, USA and Scottish dialects, Creoles, and Child English (see Drubig 2001, p.36).

1. Only the M_{ep} should bear an epistemic reading
2. Negation should take scope over M , but not the M_{ep}
3. M , but not M_{ep} should carry tense

Parallels can be drawn between these characteristics and the behaviors observed in our two Creoles



MODALS IN CREOLES

Bernabé (1983), DeGraff (1992) and Syea (2017) state that :

- When negation/tense pre-position, *pé/dwet* and *ka/dwe* is deontic
- When negation/tense post-position, *pé/dwet* and *ka/dwe* is epistemic

Epistemic modals are « higher » relative to negation and the anterior marker than deontic modals

1. Only the M_{ep} should bear an epistemic reading
2. Negation should take scope over M , but not the M_{ep}
3. M , but not M_{ep} should carry tense

1) Negation pre-position

Pyè **pa** *dwet/pé* vini

Pyè **pa** *dwe/ka* vini

Peter **not** *MUST/CAN* come

"Peter is forbidden from coming" /

"Peter cannot come"

2) Tense pre-position

Pyè **té** *dwet/pé* vini

Pyè **te** *dwe/ka* vini

Peter **ANT** *MUST/CAN* come

"Peter had to come" /

"Peter was able to come"

3) Negation post-position

Pyè *dwet/pé* **pa** vini

Pyè *dwe/ka* **pa** vini

Peter *MUST/CAN* **not** come

"It is likely that Peter is not coming" /

"It is possible that Peter doesn't come"

4) Tense post-position

Pyè *dwet/pé* **té** vini

Pyè *dwe/ka* **te** vini

Peter *MUST/CAN* **ANT** come

"It is likely that Peter came" /

"It is possible that Peter came"

MODALITY AND FINITENESS

Applying our theories of finiteness and modality:

1. Epistemic modals are raising (M_{ep} position) :
Dependent/Tensed
2. Deontic modals are exhaustive control
(M position) : Anaphorical

In original treatments and grammars, both *pé/dwet* and *ka/dwe* and both negation/tense were treated indiscriminately

No recent reanalysis of the modals to account for potential changes in patterns from the past 40 years

Important distinctions can be made if we separate them

	Jan	dwet	manjé
	John	MUST	eat
[M]	«John has to eat»		
[M_{ep}]	« <i>It is likely that John is eating</i> »		

	Jan	dwet	té	manjé	yè
	John	MUST	ANT	manjé	yesterday
[M]	#«John is obliged to have eaten yesterday »				
[M_{ep}]	« <i>It is likely that John ate yesterday</i> »				

	Jodi-a	Jan	dwet	manjé	demen
	Today	John	MUST	eat	tomorrow
	# « <i>Today John is forced to eat tomorrow</i> »				
[M_{ep}]	# <i>It is likely, today, that John is eating tomorrow</i>				

CORPUS AND ACCEPTABILITY JUDGMENTS

- **Corpus (2002 - 2022)**

- Martinican Creole : 18 MC creole authors / News articles and blog posts / ~14k types / ~280k tokens
- Haitian Creole : Wiki pages and news articles / ~42k types / ~1.7M tokens

- **Acceptability judgment survey**

- 19 MC speakers and 15 HC speakers
 - ~ 40 sentences to judge
 - Context + sentence to judge
 - Likert scale 1 - 5

1 / 4 Ki manniè ou ké jijé fraz-tala daprè konteks-la?

Kontèks: Pyè té tjit poul pou manjé oswè-a mé man pa enmen poul.

Fraz:
"Man lé pa manjé poul-la."

- ☐ 1. Fraz-la pa akseptab pies
- ☐ 2. Fraz-la pa telman akseptab
- ☐ 3. Ni yon ni lot
- ☐ 4. Fraz-la tibren akseptab
- ☐ 5. Fraz-la akseptab toubonnman

Variables	Epistemic Context	Deontic Context
Negation-MUST/CAN		X
MUST/CAN-Negation	X	X
Tense-MUST/CAN		X
MUST/CAN-Tense	X	X

RESULTS FOR MODALS IN MC

Dwet follows the patterns described in literature

- Limitations include the low number of occurrences of MC dwet altogether

Pé follows the patterns too **except for pé pa**

- Higher frequency of post-position with negation
- Post-posed negation = deontic interpretation
- If both tense and negation are present, both are preposed

	Pre-position	Post-position
Dwet	Pyè [té dwet] vini (11) Pyè [pa dwet] vini (1)	Pyè [dwet té] vini (3) Pyè [dwet pa] vini (1)
Pé	Pyè [té pé] vini (104) Pyè [pa pé] vini (17)	Pyè [pé té] vini (0) Pyè [pé pa] vini (229)
ENG	<i>Peter had to come</i> <i>Peter mustn't come</i>	<i>Peter must have come</i> <i>Peter must not be coming</i>
FR	<i>Pierre devait venir</i> <i>Pierre (ne) doit pas venir</i>	<i>Pierre doit être venu</i> <i>Pierre doit (ne) pas venir</i>

1. **Pre-position + dwet / Té pé** and **pé pa** is more acceptable in a deontic context
2. **Post-position + dwet / pé té** is more acceptable in an epistemic context
3. **Dwet té** has more variable judgments
4. **Dwet pa** in a deontic context was disfavored
5. **Pé té** in a deontic context was disfavored
6. **Pé pa** in an epistemic context has more variable judgments

Affinity between **dwet** + tense in deontic contexts vs. affinity with **pé** + negation

A NOTE ON MC **SÉ PÉ**

In corpus, MC **pé** is exclusively deontic. The only epistemic occurrences in corpus are with « sé pé ».

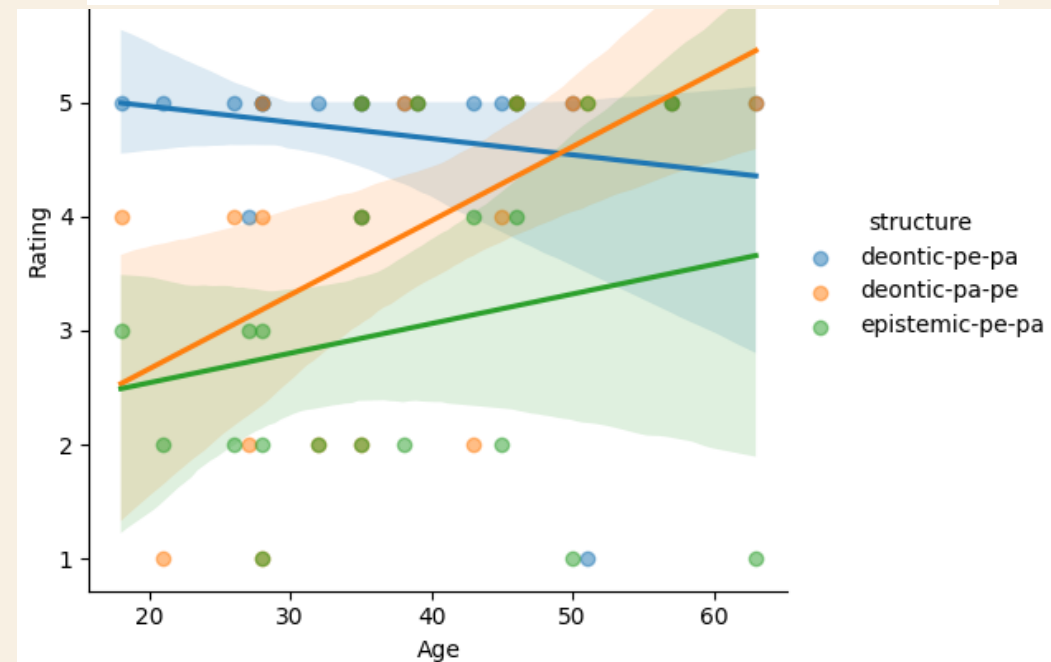
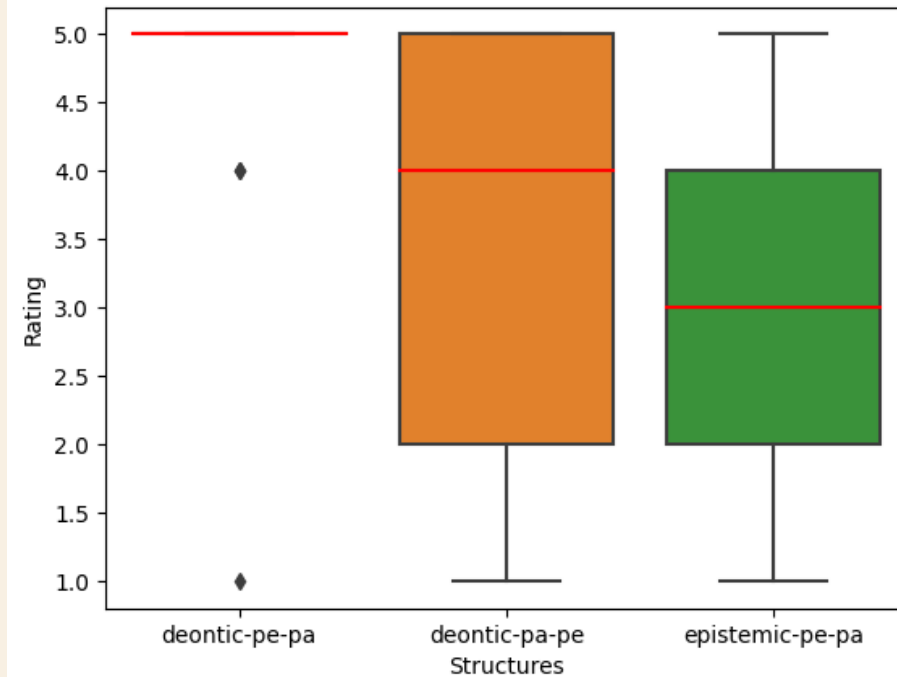
Sé pé can be both deontic and epistemic.

1. « *kisa ou vini chèché isiya, ou **pa sé pé** rété lakay ou ?* » (Barthéléry)
Who are you coming to get, couldn't you stay home?
2. « *Sézè, kon nenpòt kiles pami nou, **sé pé pa jenmen** net.* » (Barthéléry)
Césaire, like anybody among us, could have never been born.
3. « *Manman manman Rénier **sé pé té** nwè.* » (Léotin)
Rénier's grandmother could have been black
4. « *Misié **sé pé té** trapé an konjésion, lè i wè pa koté Mòn-Obé, té ni anlo pano fotovoltayik asou an tè agrikol* » (Duranty)
He might have had a stroke, when in Mòn-Obé, there were a lot of solar panol on agricultural land

MC MODAL *PÉ*

In corpus, MC *pé* is exclusively deontic. The only epistemic occurrences in corpus are with « sé pé ».

1. *pé pa* is categorically preferred in a deontic context
2. *pa pé* is more variable : older speakers tend to accept *pa pé* in a deontic context more than younger speakers
3. Younger speakers tend to prefer *pé pa* over *pa pé* in a deontic context
4. Consider language change in progress



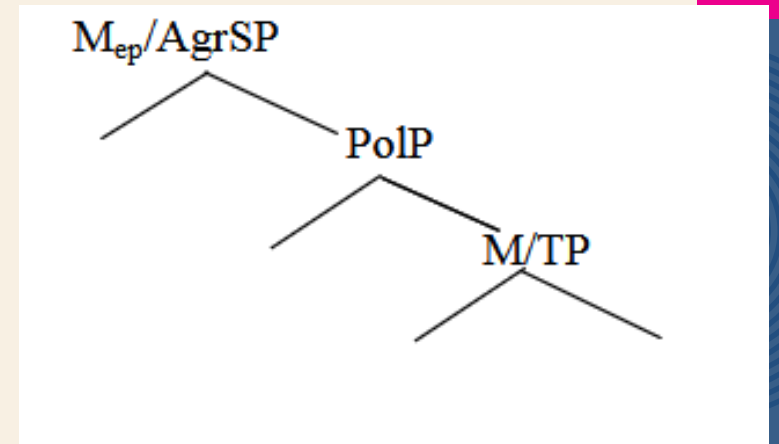
A NOTE ON MC *PÉ* + *PA*

Either the modal is in M, which means that there exists a lower position where negation can be hosted is available – but it does not explain the sentential negation scope behavior

- « Kidonk **pèsonn pé pa** di ki kréyol sé pa an bagay ki pòtalan . »
« So nobody can say that Creole is not an important thing. »
(Jean Laurent Alcide)
- « Adan konba-tala, jòdijou, **pèsonn pé pa** di kiles ki ké ni rézon. »
« In this fight, today, no one can say who will be right. »
(Serge Restog)
- « Fransé aka di tou : **Piess nonm pé pa** mété an sèviss li ni tan an, ni la maré . »
« The French also say : No man can put the time or the tide at his service »
(Thérèse léotin)

Or the modal raised to a higher position than originally and we may have evidence for verb movement in MC

More tests needed to really identify the actual position of **pé**



FINDINGS

- Work is tied to research conducted on the relevance or lack thereof of the notion of finiteness in Creoles
- Modality is one way to identify the (non-)finite properties of complements in languages and I have shown that Creole languages are worth including in the conversation
- Looking at modals solely from a morpho-syntactic viewpoint and approaching them as a single category hides the particularities of this class
 - Especially Creole-specific behaviors like MC pé pa, and pé vs. sé pé
 - MUST affinity with tense vs. CAN affinity with negation
- Look into the syntax-semantics interface when it comes to modality in both HC and MC, and look at MUST and CAN separately
- There is a need to look further into scopal relations pertaining to negation in MC and split it from the temporal specification of the complements

CONCLUSION

- Pilot of wider research on modality in Creoles and isolating languages
- Allowed me to reanalyze their behaviors in light of new theories more rigorously
- Ultimately, this allows for cross-linguistic comparisons
- Provides the tools to observe language change from a more holistic viewpoint rather than lexifier-creole paradigm

Next steps:

- Replicate the experiment by looking at more native speaker judgments on more contexts:
 - Include negation, NPI behavior, TMA marker combination, post-pre position combinations, and multiple modals
 - Look into *sé pé* vs. *pé* for MC
- Collect more diverse corpus data, as well as more spontaneous speech

THANK YOU

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